

TRAINING EFFECTS ON ENDURANCE CAPACITY IN MAXIMAL INTERMITTENT EXERCISE: COMPARISON BETWEEN CONTINUOUS AND INTERVAL TRAINING

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ABSTRACT

Tanisho, K and Hirakawa, K. Training effects on endurance capacity in maximal intermittent exercise: comparison between continuous and interval training. *J Strength Cond Res* 23(8): 2405–2410, 2009—The purpose of this study was to examine the effects of 2 different training regimens, continuous (CT) and interval (IT), on endurance capacity in maximal intermittent exercise. Eighteen lacrosse players were divided into CT ($n = 6$), IT ($n = 6$), and nontraining ($n = 6$) groups. Both training groups trained for 3 days per week for 15 weeks using bicycle ergometers. Continuous training performed continuous aerobic training for 20–25 minutes, and IT performed high-intensity pedaling comprising 10 sets of 10-second maximal pedaling with 20-second recovery periods. Maximal anaerobic power, maximal oxygen uptake ($\dot{V}O_{2\max}$), and intermittent power output were measured before and after the training period. The intermittent exercise test consisted of a set of ten 10-second maximal sprints with 40-second intervals. Maximal anaerobic power significantly increased in IT ($p \leq 0.05$), whereas $\dot{V}O_{2\max}$ increased in both training groups ($p \leq 0.05$). In the intermittent exercise test, the average of the total mean power output (1–10 sets) increased in both training groups ($p \leq 0.05$); however, the mean power output in the last stage (8–10 sets) and fatigability improved only in IT. Consequently, continuous aerobic training reduced lactate production and increased the mean power output, but there was little effect on high-power endurance capacity in maximal intermittent exercise. In contrast, although lactate production did not decrease, IT improved fatigability and mean power output in the last stage. These results indicated that the endurance capacities for maximal intermittent and continuous exercises were not identical. Ball game players should therefore improve their endurance capacity

with high-intensity intermittent exercise, and it is insufficient to assess their capacity with only $\dot{V}O_{2\max}$ or continuous exercise tests.

KEY WORDS intermittent power output, fatigability, recovery

INTRODUCTION

Most ball games, such as football, basketball, and hockey, involve periods of intermittent activity including high-intensity exercise alternating with either rest or low-intensity exercise that requires both anaerobic and aerobic energy (7). During maximal exercise, the energy is provided via the anaerobic pathway, mainly through phosphocreatine (PCr) degradation. Phosphocreatine is resynthesized using energy from oxidative metabolism during recovery (6,7,13). Quick recovery from fatigue and maintenance of high power output are indispensable for successful performance, especially for ball game players who make repeated maximal efforts with brief rest periods.

Strong correlations between aerobic fitness and endurance capacity have been reported. High aerobic fitness causes increased oxygen consumption during exercise and results in decreased utilization of energy from anaerobic glycolysis and lactic acid production (5,8). Moreover, several studies have reported that in persons with high aerobic capacity, PCr resynthesis and lactate removal are executed rapidly during rest periods (1,17,18). For high-power endurance, it is important to train for improving aerobic capacity, and therefore, continuous training (CT) has been recommended for ball game players as well.

On the other hand, some studies state that although there exists a relationship between aerobic fitness and recovery during intermittent exercise in the case of submaximal exercise, a strong correlation is rare in high-intensity exercise (3,9). Cooke et al. (3) examined the difference in the rate of metabolic recovery following high-intensity anaerobic exercise between high- and low-aerobic power groups. They found no significant differences between these groups, suggesting that the differences in recovery rate were influenced mainly by factors other than maximal oxygen uptake ($\dot{V}O_{2\max}$). It is controversial whether increases in aerobic

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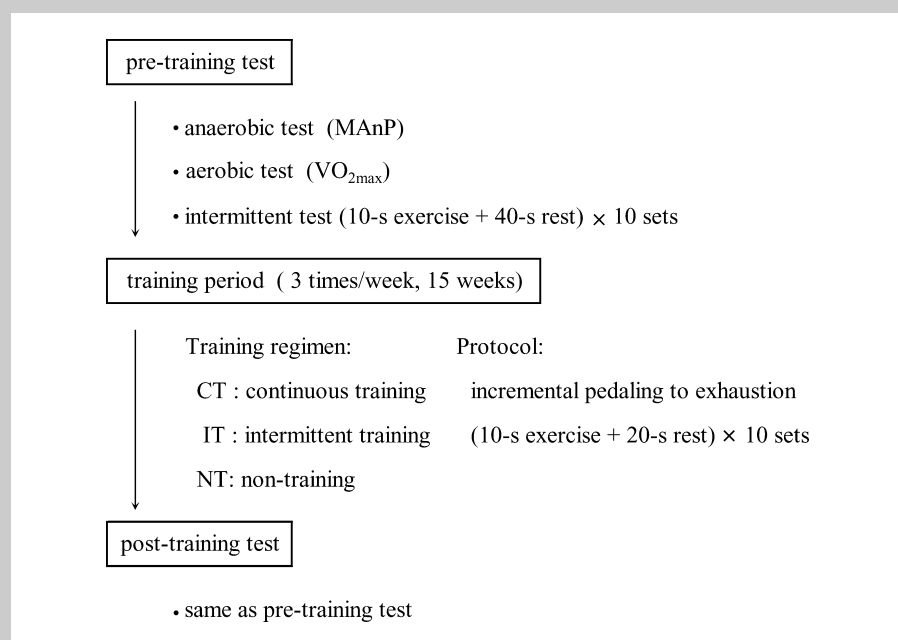


Figure 1. Experimental design. MAnP = maximal anaerobic power.

fitness with low-intensity CT improve endurance capacity specific to maximal intermittent exercise.

It is well known that interval training (IT) also improves aerobic fitness as in the case of CT (14–16), but there seems to be little information about the more suitable training method for improving endurance capacity in maximal intermittent exercise. We need to demonstrate this with actual training experiments. The purpose of this study was to examine the effects of 2 different training regimens, CT and IT, on endurance capacity specific to maximal intermittent exercise.

METHODS

Experimental Approach to the Problem

The present study was designed to compare the effects between the 2 training regimens. The experimental design is shown in Figure 1. Anaerobic and aerobic capacities were measured for all subjects, and an intermittent exercise test

was conducted as a pretraining test. Then, they were divided into 2 training groups, CT and IT, and a nontraining (NT) group. Both groups trained 3 days per week for 15 weeks. After the training period, post-training tests were conducted using the same protocol as the pretraining test, and the endurance capacities in maximal intermittent exercise were recorded.

Subjects

Eighteen male college students participated in this study. They were all competitive lacrosse players who practiced 5 days a week. The subjects were asked to refrain from any severe physical activity that might influence the test performance for 24 hours before each test. After the pretraining test, the subjects

were divided into 3 groups: CT ($n = 6$), IT ($n = 6$), and NT ($n = 6$) according to the results of the pretraining test so that there were no significant differences in maximal anaerobic power (MAnP) and $\dot{V}O_{2\max}$ between each group (Table 1).

The subjects were fully informed of all experimental procedures and the possible study-related repercussions, and informed consents were obtained. The Human Ethics Committee of Faculty of Human Development, Kobe University, approved the study.

Procedures

The following 3 tests were conducted to assess the subjects' anaerobic and aerobic capacities and intermittent power outputs before and after training. All subjects were familiarized with pedaling until fully confident of producing an all-out effort. Before the tests, each subject adjusted the seat and handle and warmed up with low-intensity pedaling at 50 W

TABLE 1. Physical characteristics of each training group.*

	CT ($n = 6$)	IT ($n = 6$)	NT ($n = 6$)
Age (y)	19.0 $\pm$ 1.1	19.3 $\pm$ 0.5	20.3 $\pm$ 1.4
Height (cm)	172.0 $\pm$ 4.5	172.2 $\pm$ 5.6	174.1 $\pm$ 3.7
Weight (kg)	62.4 $\pm$ 3.6	63.0 $\pm$ 3.5	64.4 $\pm$ 5.3
MAnP ($\text{W} \cdot \text{kg}^{-1}$)	12.5 $\pm$ 0.5	12.8 $\pm$ 0.8	12.9 $\pm$ 0.8
$\dot{V}O_{2\max}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	52.2 $\pm$ 3.5	50.8 $\pm$ 2.0	50.4 $\pm$ 4.9

*CT = continuous training group; IT = interval training group; NT = nontraining group; MAnP = maximal anaerobic power.

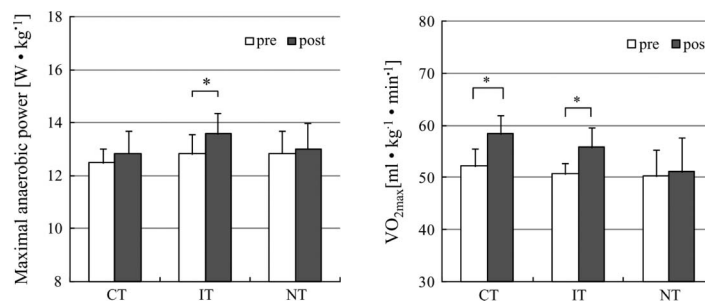


Figure 2. Comparisons of maximal anaerobic power (left) and $\dot{V}O_{2\max}$ (right) between pre- and posttraining tests. CT = continuous training group; IT = interval training group; NT = nontraining group. Asterisk indicates significant difference between pre- and posttraining tests ($p \leq 0.05$).

for 10 minutes and few seconds of high-intensity pedaling. They were instructed to remain seated during sprints and were provided verbal encouragement for maximal effort. The 3 tests were each separated by a 2-day rest period.

Anaerobic Test. Maximal anaerobic power (11) was measured to assess anaerobic capacity. Subjects executed 3 maximal pedaling sessions on a cycle ergometer (Power Max-V; Combi, Tokyo, Japan) for 10 seconds at different loads, with sufficient recovery between each session. Maximal anaerobic power ($W \cdot kg^{-1}$) was calculated from the relationship between load and power output, based on previous studies (4,11).

Aerobic Test. $\dot{V}O_{2\max}$ was measured to assess the aerobic capacity. Subjects performed the maximal incremental cycling test to exhaustion on a cycle ergometer (Aerobike 75XLIIME; Combi). The test began at a power output of 50 W, which was increased by 50 W every 3 minutes until the subject could not maintain the required pedaling rate of 60 rpm. Expired air was analyzed using the breath-by-breath method with a gas analyzing system (AE-280; Minato, Osaka, Japan). Before and after the test, the system was calibrated using gases of known concentration. The

highest $\dot{V}O_2$ measured during exercise was determined as $\dot{V}O_{2\max}$ ($ml \cdot kg^{-1} \cdot min^{-1}$).

Intermittent Exercise Test. The intermittent exercise test was performed on a cycle ergometer (Power Max-VII; Combi). The pedaling workload of each subject corresponded to the value at which MAnP was observed. The protocol comprised 10 sets of 10-second maximal sprints with 40-second recovery periods. The mean power outputs were displayed on the monitor of the cycle ergometer for each

set. Blood was collected from each subject's finger immediately after the 5th and 10th sets, and the blood lactate concentration was determined using a lactate analyzer (Lactate-Pro; Arkay, Kyoto, Japan). In this study, sets 1–3 and 8–10 were defined as early and late stages, respectively. Fatigability was calculated to assess endurance capacity.

Training Program. The training groups CT and IT were trained for 3 days per week for 15 weeks after completing the team practice. These training sessions were each separated by a 1- or 2-day rest period, and these were executed during the lacrosse season, i.e., from August to November, 2005. Each training session was performed as described below.

Continuous training groups performed aerobic exercise using a cycle ergometer (Aerobike 75XLIIME; Combi). Before starting the training, the seat and handle were adjusted for each subject, and they warmed up using the cycle

TABLE 2. Comparisons of the power outputs during intermittent exercise between the pre- and posttraining tests.*

	CT (n = 6)		IT (n = 6)		NT (n = 6)	
	Pre	Post	Pre	Post	Pre	Post
Mean power output ($W \cdot kg^{-1}$)						
Early stage (1~3 set)	11.8 ± 0.6	12.2 ± 0.6	12.6 ± 0.4	13.1 ± 0.4†	12.5 ± 0.4	11.9 ± 0.4
Late stage (8~10 set)	9.5 ± 0.6	10.0 ± 0.5	9.4 ± 0.9	10.5 ± 1.1†	9.4 ± 0.9	9.6 ± 1.1
Total (1~10 set)	10.4 ± 0.5	11.0 ± 0.3†	10.5 ± 0.6	11.5 ± 0.7†	10.5 ± 0.6	10.3 ± 0.7

*CT = continuous training group; IT = interval training group; NT = nontraining group.

†Significant differences between pre- and posttraining tests ($p \leq 0.05$).

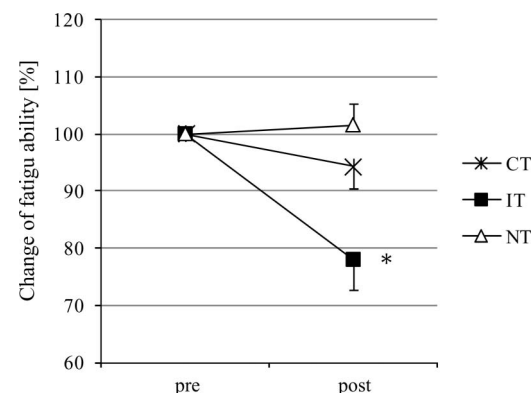


Figure 3. Change in fatigability. CT = continuous training group; IT = interval training group; NT = nontraining group. Asterisk indicates significant difference compared with the pretraining test ($p \leq 0.05$).

ergometer for 5 minutes. The initial workload was set at 120 W for 15 minutes at 60 rpm, and the average heart rate was about 140–150 $\text{b} \cdot \text{min}^{-1}$, which approximately corresponds to 70–75% HRmax. The workload was then increased by 30 W per minute until the subject could no longer maintain the required pedaling rate of 60 rpm. The exercise lasted 20–25 minutes, and the workload on exercise completion was 270–330 W.

Interval training groups performed IT using a cycle ergometer (Power Max-V; Combi). Subjects were instructed to remain seated during sprints. Before starting the training, the seat and handle were adjusted for each subject, and a sufficient warm-up consisting of low-intensity pedaling at 50 W for 5 minutes and few seconds of high-intensity pedaling was performed. The protocol of the intermittent exercise test comprised 10 sets of 10-second maximal pedaling with 20-second recovery periods. The pedaling workload was set for the value at which MAnP was observed, which was 5.3–8.4 kp in this study.

Statistical Analyses

The results were expressed as mean $\pm$ SD. An SPSS-PC statistical package (SPSS, Inc., Chicago, IL) was used to perform statistical analysis. The assessment of power output in the intermittent exercise test and the differences in the mean values between the pre- and posttraining tests were determined using repeated 2-way analysis of variance. When significant F values were found ($p \leq 0.05$), Scheffé's post hoc test was employed. Paired Student's t -test was used to compare the rate of change of fatigability and blood lactate concentration between pre- and posttraining tests. The statistical significance was set at $p \leq 0.05$.

RESULTS

The changes in MAnP and $\dot{V}O_{2\text{max}}$ are shown in Figure 2. Although MAnP increased in both training groups (CT,

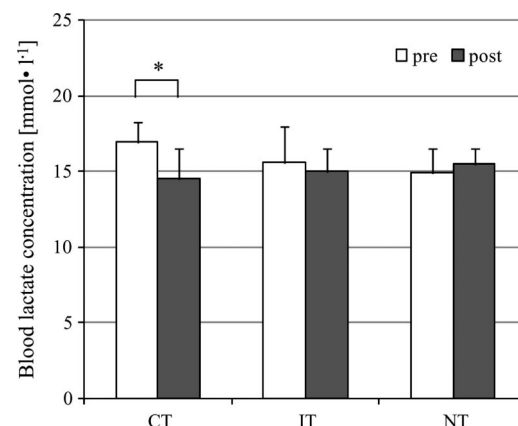


Figure 4. Blood lactate concentrations of CT (left) and IT (right) during the intermittent exercise test. CT = continuous training group; IT = interval training group; NT = nontraining group. Asterisk indicates significant difference between pre- and posttraining tests ($p \leq 0.05$).

+2.9%; IT, +6.0%), significant differences were observed only in IT. $\dot{V}O_{2\text{max}}$ increased significantly in both training groups (CT, +11.7%; IT, +9.9%). There were no significant changes in NT.

Comparisons of the power outputs during intermittent exercise between the pre- and posttraining tests are shown in Table 2. $\text{MPO}_{\text{early}}$ increased significantly in IT (+4.0%), while there were no significant changes in CT and NT (effect size = 0.69). Although MPO_{late} tended to increase in CT and IT (+5.3 and +11.7%, respectively), a significant difference was observed only in IT (effect size = 0.72). $\text{MPO}_{\text{total}}$ increased significantly in CT and IT (+5.8 and +9.5%, respectively), while there were no significant changes in NT (effect size = 0.62). The rate of change of fatigability is shown in Figure 3. It improved only in IT; there were no significant differences in CT and NT. The pre- and posttraining blood lactate concentrations are shown in Figure 4. Large accumulations were observed in both groups; more than 10 $\text{mmol} \cdot \text{L}^{-1}$ of lactate had accumulated after 5 sets. Accumulations tended to decrease in CT compared with the pretraining test, and a significant difference was observed in the mean value obtained after the 5th and 10th sets (−14.3%).

DISCUSSION

The purpose of this study was to examine the effects of 2 training regimens on endurance capacity in maximal intermittent exercise. Because of training, $\dot{V}O_{2\text{max}}$ and $\text{MPO}_{\text{total}}$ were significantly increased in both CT and IT compared with the pretraining test. No improvements were observed in NT. These results indicated that training protocols used in this study were effective in improving the performances of competitive ball game players.

Continuous training resulted in an increase in $\dot{V}O_{2\max}$ and MPO_{total} . Adaptation to aerobic training is well known. Fox et al. (6) suggested that oxygen uptake in intermittent exercise affects the aerobic energy supply during exercise and recovery periods, assuming that PCr is resynthesized more quickly during recovery by improving aerobic capacity. In this study, aerobic energy supply seemed to increase with CT, and the fatigue produced by the anaerobic pathway was reduced. This was confirmed by a significant decrease in blood lactate concentration in CT.

However, there were little changes in endurance capacity (MPO_{late} and fatigability) compared with the pretraining test. Cooke et al. (3) and Hoffman et al. (9) suspected the correlation between $\dot{V}O_{2\max}$ and recovery indices in high- or maximal-intensity exercise. Their results support our study, suggesting that endurance capacity specific to maximal intermittent exercise is not significantly affected by low-intensity CT in spite of an improvement in $\dot{V}O_{2\max}$ and reduced lactate production.

On the other hand, both MAnP and $\dot{V}O_{2\max}$ increased with IT, and all parameters in the intermittent exercise test— MPO_{early} , MPO_{late} , MPO_{total} , and fatigability—improved significantly, demonstrating that IT improves both anaerobic and aerobic fitness. Tabata et al. (16) reported that anaerobic capacity increased by 28% and $\dot{V}O_{2\max}$ increased by 7 $\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$ in high-intensity IT. Sharp (14) and Simoneau et al. (15) showed similar results. Some studies also reported that factors such as muscular buffering capacity, neural adaptation, and oxidative phosphorylation improved with maximal IT (10,12). Although lactate production did not change compared with the pretraining test in the present study, the above-mentioned improvements might enable high power output and endurance.

In both IT and CT, MPO_{total} increased significantly. However, fatigability improved only in IT, in spite of a similar increase in $\dot{V}O_{2\max}$ in both training groups. This implies that $\dot{V}O_{2\max}$ is a poor index for estimating endurance capacity specific to maximal intermittent exercise. Cooke et al. (3) and Boulay et al. (2) also suggest that maximal aerobic power does not always represent endurance performance because $\dot{V}O_{2\max}$ is limited by both central (cardiorespiratory fitness and oxygen transportation) and peripheral (oxygen utilization and transportation) factors. While CT might improve the oxidative system and reduce lactate production, muscular buffering capacity and PCr resynthesis might not be enhanced because maximal pedaling is not performed in CT and rapid recovery of PCr is not required (3,12). Consequently, CT may not improve endurance capacity specific to maximal intermittent exercise. In contrast, significant improvements might be observed with CT if intermittent exercise tests are not maximal exercises, as reported in previous studies (17,18), and do not require a large amount of anaerobic energy.

Thus, CT reduced lactate production and increased MPO_{total} . It was considered that the aerobic energy supplies

during exercise increased and that CT was effective in reducing fatigue. However, there seemed to be little effect on the rapid resynthesis of anaerobic energy supplies in maximal intermittent exercise. In contrast, IT improved the power output or endurance capacity, although lactate production did not change compared with the pretraining test. It was concluded that endurance capacity attributed to quick recovery in maximal intermittent exercise was effectively improved by IT.

PRACTICAL APPLICATIONS

Endurance capacity in maximal intermittent exercise was not improved by low-intensity CT despite a significant increase in aerobic capacity ($\dot{V}O_{2\max}$). These results indicated that endurance capacities for maximal intermittent and continuous exercises are not identical. Therefore, ball game players should improve their endurance capacity with high-intensity intermittent exercise, and it is insufficient to assess their capacity with only $\dot{V}O_{2\max}$ or continuous exercise tests.

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